



United International University

Department of Computer Science and Engineering

CSE 4495: Software Testing and Quality Assurance

Final Examination: Fall 2025

Total Marks: 40 Time: 2 hour

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. (a) i. In 2020, following a cybersecurity alert issued by the central bank regarding a potential hacking threat, several commercial banks in Bangladesh restricted ATM services during late-night hours as a precautionary measure. Although the ATM systems and banking software were operational, customers were unable to withdraw cash during these periods, leading to temporary service unavailability.
Which software/system quality attribute is primarily reflected in this scenario? Justify your answer. [CO1 – 2]
 - ii. A banking transaction processing system runs continuously in production and experiences repeated failures over time, after which it is repaired and restored to service. The engineering team wants to analyze how frequently failures occur in order to evaluate the system's long-term operational stability.
Which reliability metric should be measured to analyze this behavior and why? [CO1 – 2]
 - (b) A web-based student registration system was monitored continuously for 10 days during the peak enrollment period. During this time, the following events were recorded:
 - The system served a total of 5,000 requests.
 - Among these, 120 resulted in incorrect registration outcomes.
 - The system crashed 20 times, requiring a restart each time.
 - Each crash caused the system to remain unavailable for 15 minutes before service was restored.
 - i. **Calculate the ROCOF (Rate of Occurrence of Failure) of the system.** [CO1 – 2]
 - ii. **Calculate the Availability of the system over the observed period.** [CO1 – 2]
 - (c) In the context of software correctness, explain the difference between the **rationalist** and **empiricist** approaches. [CO1 – 2]
2. (a) A large-scale ride-sharing and food delivery platform **UIU-Go** evaluates each customer request using the following **given choices and representative values**:

Choice	Representative Values
Service Type	Ride, Food Delivery, Book Rental
Request Timing	Instant, Scheduled
Partner Availability	Available, Busy
Item Availability	Available, Will be available, Not Available
Customer Tier	Regular, Premium
Payment Method	Cash, Digital Wallet
Promo Code Status	Valid, Invalid, Not Used

The platform enforces the following business rules: a request cannot be fulfilled if the partner (and/or item) is busy, regardless of other conditions. Scheduled requests are supported only for ride services and not for food delivery. Promo codes are applicable only for food delivery and only when payment is made using a digital wallet, while an invalid promo code always results in an error. When surge pricing is active, cash payments are not allowed. Premium customers are allowed to access restricted zones, whereas regular customers are not. Promo codes are optional and may remain unused even when applicable.

Using the Category Partition Method (CPM), calculate the total number of test specifications before & after applying any constraints. [CO3 – 7]

- (b) When is the **Category Partition Method (CPM)** not suitable for reducing test design, and what testing technique should be used instead? Illustrate your answer with a brief example. [CO2 – 3]

3. You are writing unit tests using **JUnit** for a utility class that contains the following methods:

```
boolean isValidUsername(String username)
void processLargeOrder()
```

The method `isValidUsername` returns `true` if the username length is at least 6 characters and contains only alphanumeric characters.

The method `processLargeOrder()` performs a time-consuming operation and is expected to complete within **500 milliseconds**.

- (a) Write JUnit test case(s) to test the `isValidUsername` method. [CO3 – 4]
(b) Write a JUnit test case to verify that `processLargeOrder()` completes within the expected time limit. Also, capture and print the actual execution time of the method. [CO3 – 5]

4. Consider the following Java method:

```
public int countEligibleItems(int[] items, int threshold) {
    if (items == null || threshold <= 0) {
        return -1;
    }

    int i = 0;
    int count = 0;

    while (i < items.length) {
        if (items[i] >= threshold) {
            count++;
        }
        i++;
    }

    return count;
}
```

- (a) Draw the Control Flow Graph (CFG) for the given method. [CO4 – 4]
(b) Design a minimal set of test cases that achieves *branch coverage*. Clearly specify the input values used in each test case. [CO4 – 4]
(c) Does your test suite achieve *statement coverage*? Justify your answer briefly. [CO4 – 3]